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Key technologies for smart energy systems: Recent developments, challenges, and research opportunities in the context of carbon neutrality

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ABSTRACT

Energy crisis and environmental pollution have expedited the transition of the energy system. Global use of low-carbon energy has increased from 1:6.16 to 1:5.37. Smart energy systems have received significant support and development to accelerate the development of smart cities and achieve the carbon neutrality goal. As a result of analyzing recent related publications and weighing their merits and downsides, it is determined that a more comprehensive and objective analysis of the main technologies underlying smart energy systems is necessary for the context of the new era. We report thoroughly on the six important hotspot technologies in smart energy systems, and undertake a comparative analysis of each technology's performance features, application status, and challenges. The ten key obstacles to developing a smart energy system are examined in terms of technology constraints, the social environment, human survival, and extreme climate circumstances. The four opportunities that smart energy systems confront in the digital age are examined in terms of governmental support, technology innovation, and public participation. The purpose of this research is to advance the creation of smart energy systems and the sustainable development of society in two ways: i) Smart energy system research should begin with a combination of technological innovation and practical application; ii) Key technologies in smart energy systems should consider the needs of people's livelihoods to evolve in a more intelligent and diverse path.

1. Introduction

Climate change, environmental pollution, energy crisis and the outbreak of COVID-19 have aroused global concern on energy use. To meet the global carbon neutrality target and resolve the contradiction between energy use and environmental pollution, all countries are aggressively developing renewable energy (RE) (Gungor and Dincer, 2021) and researching the integrated use of multiple energy sources (Ahmad and Zhang, 2020). When the energy industry undergoes dramatic transformations, information technology advances at a breakneck pace (Heffernan, 2012). Thus, comprehensive integration of new energy and information technologies, as well as the establishment of a highly intelligent, information-transparent, open and connected smart multi-energy system (MES), are critical for increasing global energy conservation and emission reduction (Mahmud et al., 2020).

Climate change is a severe issue that humanity must address. By 2020, the United Nations Framework Convention on Climate Change requires parties to adopt a long-term strategy for low-carbon development (United nations, 1992). As a result, governments have committed to carbon neutrality since 2018. Additionally, the energy production structure has evolved due to ongoing efforts to improve energy conservation and emission reduction. The use of conventional fossil fuels has been phased out in favor of various low-carbon energy sources. The global energy structure and the power energy structure are depicted in Fig. 1 (a) and (b), respectively. In 2019, global low-carbon energy production totaled 25,428 TWh, or 15.7 percent of total energy production, and it is used to generate 9785.77 TWh of electricity, or 36.7 percent of total electricity generation. By 2020, total energy production from low-carbon energy reached 10,109.12 TWh, accounting for 39.1 percent of total energy and representing a 3.3 percent increase over

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2019 (Our World in Data, 2019a). The global status of low-carbon energy power generation is depicted in Fig. 1 (c). Several countries, including Sweden, France, and Norway, have achieved essentially 100% low-carbon energy generation. Certain countries, including China, the United States of America, Russia, and Japan, have achieved between 30% and 40% low-carbon energy power generation.

While the global energy production structure has changed, the global energy consumption structure has also changed (Azadeh and Tarverdian, 2007). Fig. 1 (d) describes the changes in the energy consumption structure during the nearly 20 years from 1999 to 2019. The changing trend of the figure shows that energy consumption is gradually transitioning from traditional fossil energy to new low-carbon energy. Especially, low-carbon energy consumption increased from 14,805 TWh in 1999 to 24,127 TWh in 2019, with a growth rate of 62.97%. While for traditional fossil energy, it increased from 91,968 TWh in 1999 to 136, 761 in 2019, with a growth rate of 48.71%. Obviously, the global consumption of low-carbon energy is growing faster than the fossil energy consumption (Kone and Buke, 2010). Therefore, it is inevitable to further develop various forms of RE power generation and multi-energy comprehensive utilization technologies in the future to realize clean energy with a high proportion of RE.

In recent years, the power industry has accelerated the development of highly flexible distributed energy, which can effectively address the issues such as serious environmental pollution, long transmission distances, and significant energy loss associated with traditional large-scale centralized power plans (Mengelkamp et al., 2018). In this context, the integrated energy system (IES), which integrates energy production, transportation, conversion, storage, and consumption, has been highly valued by academia and industry (Cao et al., 2018). IES plays a highly promoting role in improving the flexibility and economy of the power grid and realizing the cascade utilization of multiple energy sources. IES can not only improve energy production and consumption methods, but also solve the problems of high volatility, randomness, and low energy flow density of RE. To this end, countries worldwide are actively exploring innovative energy systems with the characteristics of multiple energy complementation, supply and demand interaction, centralized and decentralized coordinated operation. Moreover, many countries have carried out related research projects, as shown in Table 1. These studies have accelerated the transition of energy consumption, and promoted the creation of a modern energy system that is clean, low-carbon, safe, and efficient.

While large-scale distributed new energy continues to be integrated into the grid, it brings new challenges such as energy waste, variable power quality, uncoordinated system operation, and low energy efficiency. The advent of the digital era has facilitated the resolution of these issues on a technical level. The history of artificial intelligence (AI) technology dates back to the mid-twentieth century, with the Turing test in 1950, the computer "Dark Blue" in 1997, and then AlphaGo in 2016. Each time it was discovered, it sparked a research frenzy (Cheng and Yu, 2019). AI technology emulates human logical reasoning and advanced intelligence through computers and continues to advance in the directions of computational intelligence, perceptual intelligence, and cognitive intelligence, thereby effectively and comprehensively assisting human work (Zhang and Lu, 2021).

The new generation of AI technology, which made significant advances in data resolution, learning capability, and processing capability, has been steadily applied to diverse spheres of society (Rozynek et al., 2021). Therefore, the development of AI technology has made the energy industry more digitized and gave birth to smart energy systems (SES). Through the incorporation of critical technologies such as intelligent control (Shahzad et al., 2020), SES can perform real-time online analysis and integration of data from multiple sources, including the mutual coupling, conversion, and feedback between various energy flows, information flows, and capital flows in the system, thus effectively

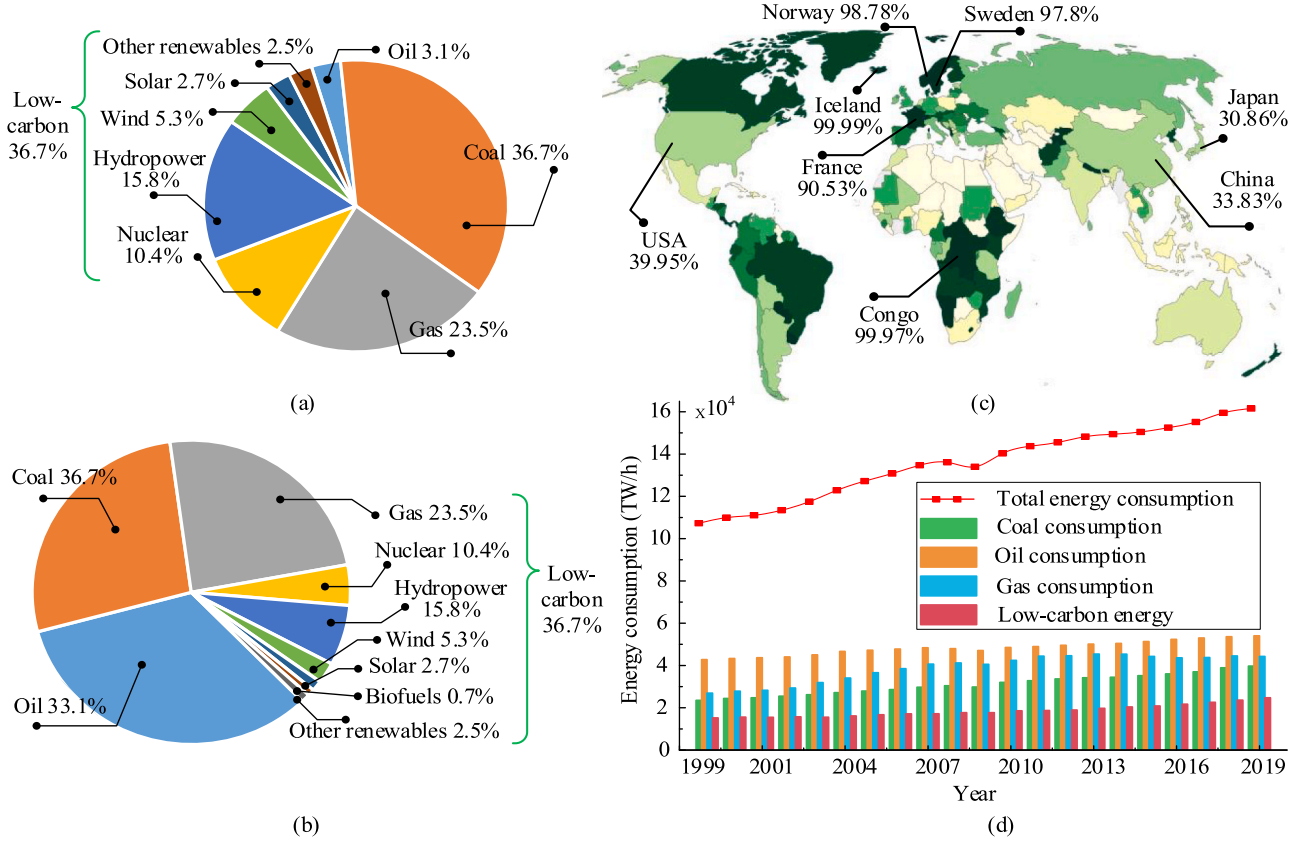


Fig. 1. Characteristics of global energy structure: (a) Global power generation energy characteristics, (b) Global energy Characteristics, (c) Global low-carbon energy power generation, (d) Global energy consumption characteristics.

Table 1
The evolution of IESs in recent years in various countries worldwide.

Years	Country	Policy/Project	Main content
2003	Switzerland	Future energy network vision (Fischler, 2012).	Propose that the future energy network is the result of the coupling of multiple energy sources.
2007	USA	Energy independence and security act (Tian, 2015).	Clearly required the electricity-gas network to conduct joint planning.
2008	USA	Future RE transmission, distribution, and management system (Shen et al., 2012).	Carried out research on energy internet of power distribution system.
2008	Germany	Information energy promotion plan (Zha et al., 2014).	Aims to promote efficient energy system project based on information and communication technology.
2009	Canada	National IES research report (Pasquale, 2012).	List the regional IES technology research and engineering construction as the national energy strategy from 2010 to 2050.
2011	Europe	Future smart energy internet project (You and Song, 2017).	Build a future energy internet information and communication platform to support the development of innovative services.
2014	Denmark	The project of EnergyLab Nordhavn (Hongbo et al., 2017).	Including multi-energy storage, electric cars, smart building, combined heat and power, and 40,000 residents, etc.
2014	Japan	IES project in Toyosu Pier, Tokyo (Mengelkamp et al., 2018).	Including gas and photovoltaic power generation, covering multiple energy forms.
2016	China	Guiding opinions on promoting the development of "Internet +" smart energy (Zhou et al., 2016).	Utilize the complementary characteristics of multi-energy to realize the use of energy ladders and promote the consumption of new energy.
2017	USA	Brooklyn microgrid project (Pan et al., 2016).	Build a peer-to-peer energy trading platform through blockchain technology.
2017	China	New energy microgrid demonstration project (Guo et al., 2017).	Promoted the first batch of 23 multi-energy complementary integration optimization demonstration projects.

achieving the coordination and complementarity between the various entities in the SES, and promote the optimal operation of the system (Tan et al., 2020). SESs have evolved steadily in recent years because of advances in a variety of fundamental and cross-border technologies. It establishes stricter rules for resource allocation across multiple energy sources and strives to ensure the system's efficiency and economy of operation, as well as the flexibility of diverse commercial applications that rely on the grid. Therefore, to highlight the necessity and value of this research, the paper comprehensively analyzed and compared the advantages and disadvantages of current review articles on SES according ascending order of the publication year. The detailed analysis content is shown in Table 2.

To summarize, a review and comparative study of the existing literature reveals that several researches are examining the concept, technology, and application of SESs. However, in the new era, significant advancements in various critical technologies have presented SESs with new opportunities and challenges. Thus, the current literatures include shortcomings such as inadequate conceptions, discussions, background, and research. To address these issues and keep up with the times, this paper comprehensively introduces the performance characteristics and application status of the six key hotpot technologies in the

Table 2
Comparative analysis of review literatures on SESs.

Ref. (year)	Main contents	Advantage and disadvantage analysis
Lund et al. (2012)	It summarizes the transition from smart grid to SES from the perspective of market and operation.	It aims to explore the early formation of SESs. With the continuous breakthrough of smart technology, key technologies in this field need to be further explored.
Mathiesen et al. (2015)	A review of how to adopt a SES approach to solve the problem of RE integration.	It aims to explore the key technology of new energy grid connection, without discussing the development challenges of other key technologies.
(Zhou et al., 2016)	A comprehensive overview of the smart home energy management system.	It aims to summarize the SES in residential areas and does not analyze the key technology of SES in other application scenarios.
Nguyen et al. (2017)	It summarizes the modeling and simulation problems of SES physics, information, application, etc.	It aims to explore specific modeling and simulation technologies and does not involve new opportunities and challenges in the construction of SESs in the context of carbon neutrality.
Lund et al. (2017)	It summarizes the scientific literature in the field of conceptual understanding of SESs.	It aims to discuss conceptual definitions and understanding issues but does not fully explore the application of intelligent technologies in SESs.
Mohammadi et al. (2018)	An overview of the benefits of the energy hub (EH) modules through the introduction of smart technology.	It aims to explore the optimal management of EHs after the introduction of intelligent technologies, but lacks more systematic research on key technologies of SESs.
O'Dwyer et al. (2019)	It summarizes the technology and implementation examples of the main application areas of SES.	It is primarily focused on the coordination and administration of SESs in the context of smart cities and does not include the development of key technologies for SESs in the context of carbon neutrality.
Xu et al. (2020)	Systematic introduction to the design, management, and optimized operation of typical SESs.	It mainly focuses on the key technology of SES optimization management and operation, but lacks discussion on other key technologies.
Rae et al. (2020)	A review of the latest research based on smart local energy systems.	It aims to discuss the key obstacle areas constituted by multiple technologies in the system, but the progress of the technology itself is insufficiently discussed.
Schweiger et al. (2020)	It summarizes the steps and obstacles of the SES business model from the perspective of consumers.	It aims to explore the user-centric system business model and does not fully involve the discussion of other key technologies.
ALiero et al. (2021)	The current trends and future challenges of smart home energy management systems are reviewed.	It aims to explore the energy consumption of autonomous devices such as smart home appliances and sensors but lacks discussion on other key technologies of SESs.
Kim et al. (2021)	A review of the transition process from smart home to smart city energy management.	It is primarily intended to assist in the development of energy-saving plans for smart cities, rather than an in-depth examination of the essential

(*continued on next page*)

Table 2 (*continued*)

Ref. (year)	Main contents	Advantage and disadvantage analysis
		technologies underlying SESs in a variety of scenarios.
Kourgiozou et al. (2021)	It summarizes the current research status of SESs in campus buildings and puts forward prospects from the perspective of policy.	It mainly focuses on areas with special campuses, and does not discuss general and universal multi-energy system technologies.
(Ahmad et al., 2021)	It summarizes the application of Internet of Things technology in 4 areas of SESs.	It is mainly focused on the research of Internet of Things technology, but the application of other intelligent technologies in SESs is not discussed, and the fields involved are limited.

SES, including new energy generation prediction technology, demand response (DR) technology, collaborative energy management of multiple energy flows, advanced energy storage (ES) technology, information exchange technology, and digital energy integration market and service mechanism. Then, the ten fundamental obstacles confronting the development of SES essential technologies are examined in terms of technical constraints, societal context, human survival, and adverse climates, promoting research on bottleneck technologies. Finally, the four major future potentials for SESs are analyzed in terms of policy support, technological innovation, and universal involvement, effectively addressing the current literature's and research's scarcity of depth and sophistication. Additionally, the paper serves as a vital reference for implementation and building the future social-economic system, effectively promoting human society's sustainable growth. Fig. 2 depicts the study's research flow chart.

2. Concept of smart energy system

With the continuous advancement of AI technology, the control and management of MESs is also developing in the direction of intelligence. In order to adapt to the development of the times and welcome the arrival of smart cities, SESs use a variety of smart calculation and measurement methods, such as smart monitoring of energy information, smart collection, analysis and processing of data, etc. (Onile et al., 2021), to achieve the comprehensive perception of energy supply, effective processing of information transmission, flexible application of energy resource, and real-time coordination of system operations (Lin et al., 2020). Intelligent control continues to integrate technologies such as fuzzy logic, big data, and deep learning to perform various forms of power generation prediction, load prediction, and uncertainty prediction for various distributed energy sources (Zhao et al., 2019). All of this provides strong support for distributed decision-making, collaborative autonomy, security protection, transaction supervision and integrated services of the MES (Kermani et al., 2021).

A vertical interactive structure of the intelligent control and management of SESs is shown in Fig. 3. The top layer is the intelligent control end, the bottom layer is the actual operation end of the MES, and the middle layer is the realization of data processing and information transmission (Ren et al., 2021). The overall operating characteristics of the SES are as follows: through the data center, the system monitors and collects real-time operating data of multiple links from the source, network, load, and storage at the operating end, and extracts and processes the collected data through big data analysis technology. Then the effective data information is transmitted to the intelligent control terminal. At this time, the intelligent control terminal adjusts the system operation strategy through this information, thus realizing the interconnection and human-computer interaction between various links in the system. The SES can effectively improve the flexibility and reliability of energy consumption, promote the consumption of RE, reduce the operating cost, promote energy conservation and emission reduction, and improve the social environmental protection benefits. In addition, compared with the operating characteristics of traditional MESs, the operation and management strategies of SESs have better integrity and encapsulation (Jia et al., 2021). For example, when the working scene and operating characteristics of the system change, the AI-based control

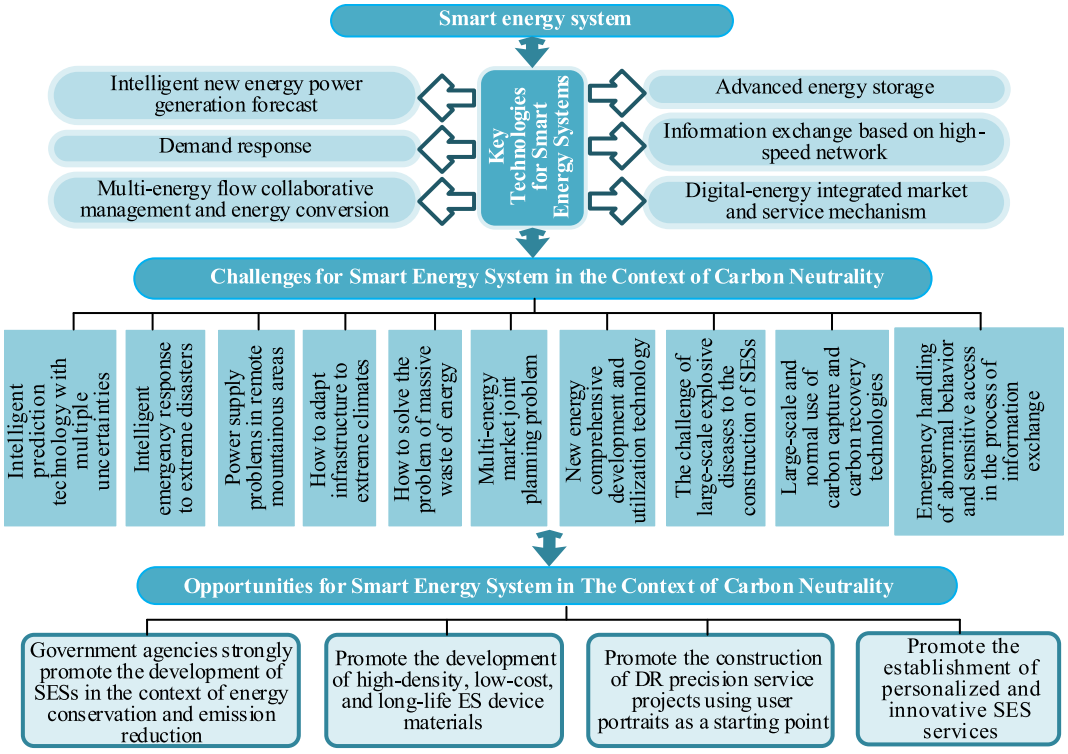


Fig. 2. A research flow chart for this study.

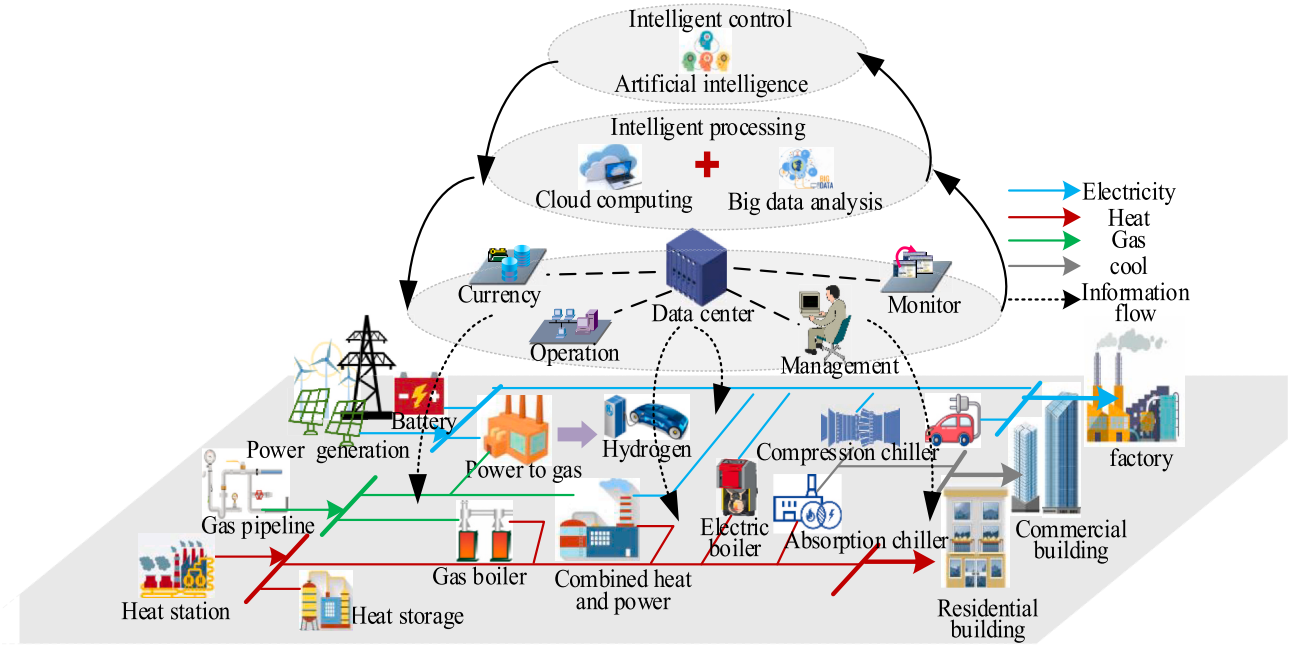


Fig. 3. Smart energy system that incorporates a variety of key technologies.

technology only needs to input the current operating state to obtain the corresponding control strategy, hence omitting the complicated constraints in the traditional mode, and making the operation of the system more efficient and concise.

3. Development of key technologies for smart energy systems

3.1. Intelligent new energy power generation prediction technology

With the advantages of clean and low-carbon, new energy has emerged as a critical driver in accelerating the energy transition and achieving global zero-carbon emissions. New energy sources have developed rapidly in recent years due to policy support and significant

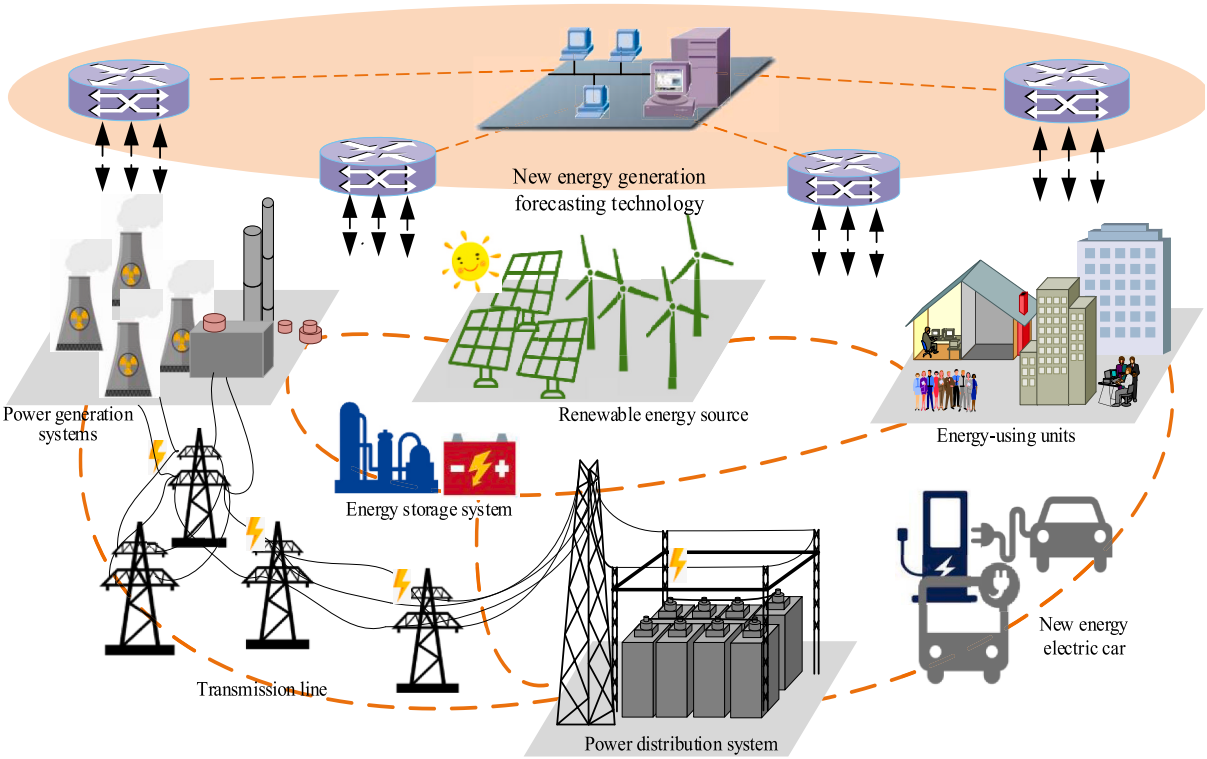


Fig. 4. Intelligent new energy power generation prediction technology of SESs.

cost reduction in wind and solar power generation. By the end of 2019, the global wind turbine installed capacity exceeded 651 GW, while the newly installed capacity was over 6000 GW (Pryor and Barthelmie, 2021). In addition, according to statistical figures (Our World in Data, 2019b), global RE power generation reached 7027.73 TWh in 2020, with hydrogen power accounting for 4222.21 TWh and wind and solar power accounting for 1429.62 TWh and 724.09 TWh, respectively.

With the rapid development of new energy power generation, its uncertainty also brings new challenges to the reliable operation of the power grid (Gao et al., 2019). For this reason, new energy power generation prediction technology has been widely studied and applied because it can alleviate this problem. Reliable prediction of new energy generation provides data for grid operation planning, which can effectively address the issue of large-scale new energy generation's volatility and anti-peaking characteristics (Hossain et al., 2021). The implementation characteristics of this technology in the SES are shown in Fig. 4. This technology collects historical data of a variety of new energy power generation through smart equipment, and uses advanced intelligent technology to extract and analyze effective data to predict the amount of power generation in the following period, thus realizing the reliable operation of various links in the system such as power generation, transmission, distribution, and power consumption. Through investigation, it is found that existing prediction methods can be classified into physical methods, statistical methods, and AI methods, depending on the models used (Wang et al., 2019). The physical method is based on numerical weather prediction, which mainly used for atmospheric dynamics forecasting. However, it takes a significant amount of processing power and is therefore unsuitable for short-term forecasting (Wu et al., 2018). The statistical method establishes a functional model of historical data and prediction objects. It aims to reveal the mathematical relationship between energy online time-series data (Hu et al., 2018), but its linear model constraints its capability to handle longer-term prediction problems. Predictive methods powered by AI offer a greater potential for data mining and feature extraction (Krawczyk et al., 2017), which broadens their application opportunities. They encompass both predictions using a shallow model and deep learning. The shallow model's feature selection process is inefficient, and its generalization capacity is limited, making it difficult to handle complicated datasets. As a result, the prediction technology based on deep learning has been further studied to solve these problems.

At present, new energy power generation prediction technology has been widely used in photovoltaics, wind energy and other fields. This technology can achieve hourly, minute, and second-level prediction performance, and plays a critical supporting role in new energy generation's safe and stable operation. Table 3 describes the application characteristics of different prediction methods. In the SES, a fast and accurate new energy generation prediction technology can promote the coordinated and stable operation of the internal source-network-load-storage sectors of the system. However, although the current prediction technology is developing rapidly, it still faces some problems in its application. On the one hand, in some extreme situations, it is difficult for the prediction technology to grasp the volatility of new energy power generation in real time, which may cause a power outage to the power grid. On the other hand, new energy power generation is widely distributed, and implementing more accurate prediction technology in some remote areas with underdeveloped intelligent technology will be an important content of the development of SESs. Therefore, future prediction technologies should follow the trend of the times, develop more innovative and advanced intelligent analysis technologies, and achieve comprehensive coverage of widely distributed new energy power generation forecasts, thereby promoting the development and implementation of SESs.

3.2. Intelligent demand response

In large-scale distributed new energy access, it is critical to

Table 3
New energy power generation forecasting methods.

Prediction technology	Main prediction method		Application characteristic
Physical method	Probability prediction of solar irradiance (Bakker et al., 2019).		It applied statistical post-processing techniques to determine relationships between observations of global radiation and forecasts of various meteorological variables.
	probability prediction of wind (Wang et al., 2017).		It proposed a method searching for the similar wind characteristics in the numerical weather predictions data.
Statistical method	Autoregressive moving average mode (Singh ET AL., 2019).		It proposed a new Repeated WT model to improve accuracy for very short-term wind speed forecasting.
	Bayesian method (Panamtash et al., 2020).		It proposed a copula-based Bayesian approach to improve probabilistic solar power forecasting by capturing the joint distribution between solar power and ambient temperature.
	Kalman filter (Yang, 2019).		It proposed two simple remedies to address the forecast horizon problem, but the effectiveness is minimal.
	Exponential smoothing method (Cheng and Li, 2011).		It proposed a forecasting model with a novel fuzzy smoothing method to deal with uncertainty appropriately.
	Markov chain model (Sanjari and Gooi, 2016).		It presented a method to forecast the generated power of PV systems based on the higher order Markov chain.
	Grey theory (Wu et al., 2018).		It proposed a novel multi-variable grey forecasting model to forecast the electricity consumption in Shandong.
Artificial intelligence method	Shallow model	Support vector machine (Ayeleru et al., 2021).	It aimed to forecast MSW with the projection period in continuing guesstimates by using supported vector machine.
		Artificial neural network (Wang et al., 2019).	It proposed a new approach to successfully combine a data preprocessing technique with a linear combination method.
		Extreme learning machine (Hossain et al., 2017).	It proposed 1 h ahead mean PV output power forecasting model based on extreme learning machine approach.
		Adaptive fuzzy network (Ghoushchi et al., 2021).	It employed an extended fuzzy wavelet neural network to predict a wind power plant's power output using weather and power plant parameters.
	Deep learning	Deep random structure (Zhang et al., 2015).	It presented a sophisticated deep-learning technique for short-term and long-term wind speed forecast.

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Table 3 (*continued*)

Prediction technology	Main prediction method	Application characteristic
	Stacking an extreme learning machine (Luo et al., 2018).	It proposed a more practical approach through the combination of extreme-learning machine method and deep-learning model to forecast wind speed.
	Machine learning (Li et al., 2018).	It developed a multi-layer artificial neural network and deep machine learning to forecast short-term wave forces.
	Deep convolutional network (Wang et al., 2017).	It proposed a novel hybrid method for deterministic PV power forecasting based on wavelet transform and deep convolutional neural network.
	Deep recursive network (Rahman et al., 2018).	It optimized novel deep recurrent neural network to long term electric load prediction at 1-h resolution
	Deep belief network (Wang et al., 2016).	It proposed a novel deep learning based approach for deterministic and probabilistic WSF to forcast wind farm.
	Stacked autoencoder (Huang et al., 2019).	It proposed a deep spatial-temporal data-driven model to predict and detect power system security weak spots.

strengthen the pleasant interaction between provider and demander to react to market demand and increase network operation efficiency and service level (da Fonseca et al., 2021). In recent years, DR technology has advanced at a fast pace. Traditional DR is classified into two types: Price-Based DR (PBDR) (Wang et al., 2017) and Incentive-Based DR (IBDR) (Zhong et al., 2012). We further subdivide it into DR based on load reduction transfer, load substitution conversion, contract signing, and market interactive control according to the specific response form and response attributes. We have used graphs to express various DR response modes and characteristics more intuitively and in detail, as shown in Fig. 5 (a) and (b).

PBDR based on load reduction and transfer means that the user realizes the load adjustment in the time dimension according to the change of the price signal (Yu et al., 2018). Users only chooses to participate in the response, and cannot convey their energy preference, so its communication is unidirectional according to the energy price signal. With the in-depth study of the coupling characteristics of multi-energy sources, users can choose different energy use forms to meet the energy demand with the same quality. Therefore, PBDR based on the substitution of different energy sources has been developed (Kok et al., 2016), allowing users to participate in DR according to their energy consumption preferences, and it has a bidirectional communication method.

IBDR based on contract agreement is realized by signing a contract agreement with the users (Rahimi and Ipakchi, 2010). The system administrator uniformly operates the load equipment of contract users, and it has a certain mandatory and unidirectional communication. With the development of the multi-energy market and the gradual maturity of the transaction mechanism, the market-oriented IBDR project has been further developed (Hassan et al., 2019). IBDR based on market interactive control strengthens the interactive capabilities between users and other entities in the market. Compared with IBDR based on the contract agreement, IBDR based on market interactive control can realize bidirectional communication, so that users are the receiver of market price strategy, and can influence the formulation of these strategies with their preference (Hamwi et al., 2021).

As the market's catalyst, intelligent DR is a concrete measure for achieving coordinated optimization of diverse energy sources and loads participation in the energy market (Lv et al., 2021). It can influence capital flow by transmitting information flow and controlling energy flow (Jiang et al., 2020). It depends on an intelligent energy information management system to analyze real-time market information (Kaluthanthrige and Rajapakse, 2021). Fig. 5 (c) shows an integrated DR interactive control framework subject to macro-control and supervision by the government. This framework enables simultaneous bidirectional adjustment between service entities and consumers, ensuring optimal resource coordination between the supplier and demander (Mnih et al., 2015). Simultaneously, it has aided in constructing and developing a multi-energy market characterized by fair competition and affordable costs. To this end, based on different DR modes, many countries have established DR demonstration projects to tap the flexibility of load-side resources fully. Table 4 summarizes the typical DR application projects in various countries.

In addition, the frequent usage of massive amounts of data and real-time decision-making make DR extremely complicated. To address these difficulties and generating optimal decisions, intelligent approaches such as machine learning can be used to build a complete set of perceptual decision-making systems via "trial and error" (Ma et al., 2020), which has been widely used in DR projects (Antonopoulos et al., 2020). For example, a DR management system based on reinforcement learning (RL) and fuzzy inference was proposed (Alfaverh et al., 2020), which can achieve the optimal operating state of smart homes under the conditions of considering user preferences. A discrete manufacturing DR scheme based on multi-agent deep RL was proposed (Lu et al., 2020), which effectively explored the important impact of the implementation of DR on power grid reliability while maintaining production tasks in the industrial sector. Weber-Fechner's law and clustering algorithm was used to construct a quantitative DR characteristic model of an electro-heating system (Zhong et al., 2021), which explored the quantification of user behavior of DR participants and the differences between users, thus providing a more intelligent system for electric heating equipment. A deep learning approach was used to explore DR projects in uncertain environments (Wen et al., 2020), which aim to formulate short-term DR plans for peak electricity demand periods and alleviate the imbalance between supply and demand in the power system. A model-free deep RL to optimize smart grid DR management was designed to solve the problem of DR that is difficult to adapt to actual operating conditions (Wang et al., 2020). An optimal DR strategy for multi-regional building HVAC systems using the supervised learning method was proposed (Kim, 2020), which achieves both personnel comfort and HVAC operation economics, so that the building's heat capacity can be effectively utilized.

At present, there are two main problems within DR technology. On the one hand, the load analysis on the user side is not accurate enough. The strong randomness on the user side and the interaction effects between multiple users have not been fully considered. Therefore, it is imperative to deeply study the uncertainty of demand brought by price sensitivity and user interaction characteristics to implement precise DR strategies. On the other hand, the current price mechanism of DR is not perfect enough, and the research on multi-energy price mechanisms is not in-depth. Price changes will change the way or frequency of energy utilization, so that the prices of different energy sources affect each other. Therefore, a sound DR price mechanism system is of great significance to the implementation of DR projects.

3.3. Multi energy flow collaborative management and energy conversion

As seen in Fig. 6, the multi-energy coupling system transforms the energy utilization from the traditional "single energy" mode to the

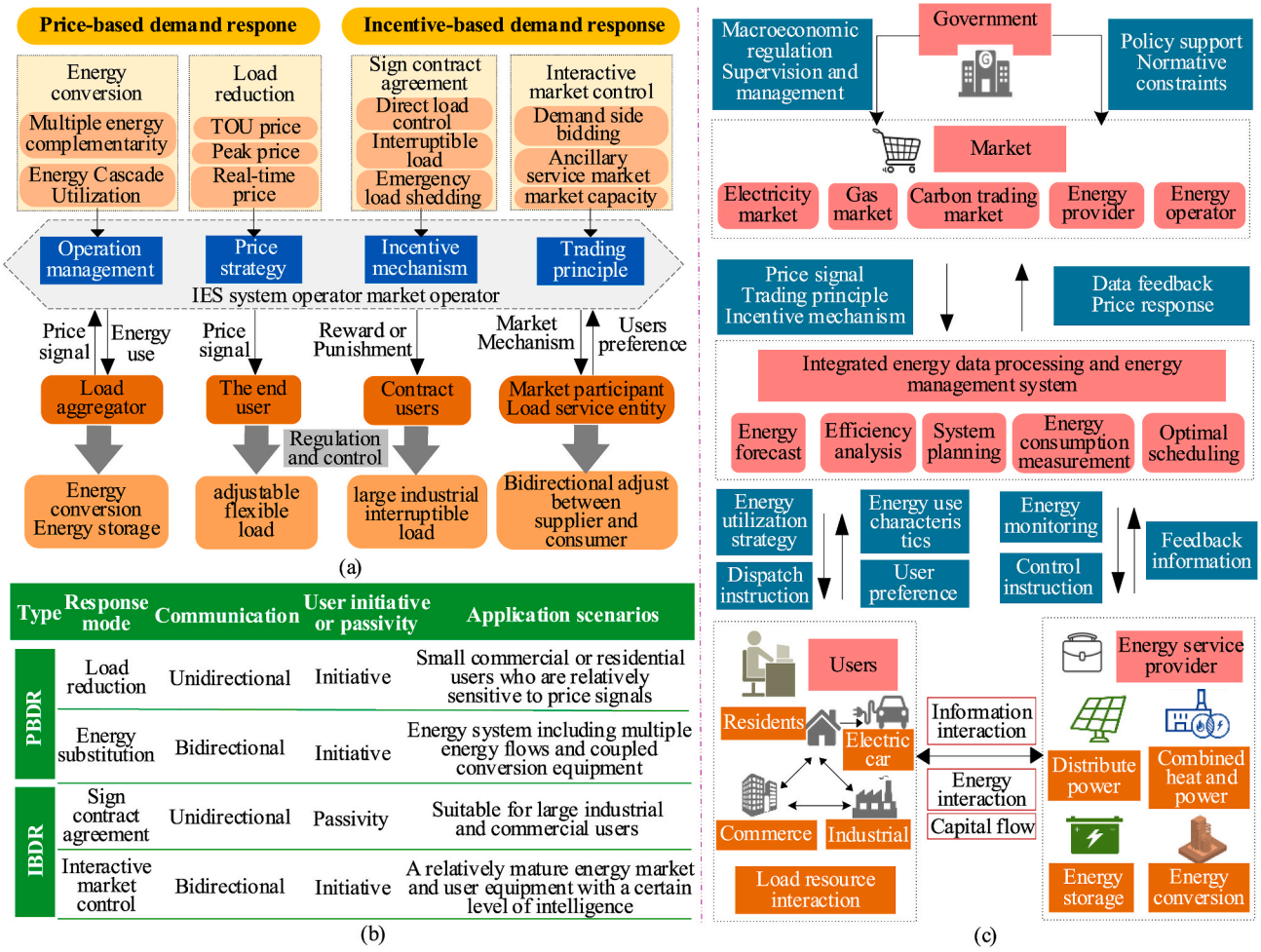


Fig. 5. Intelligent demand response technology: (a) Different demand response modes, (b) Different characteristics of different demand response modes, (c) Intelligent demand response interaction framework.

"multi-energy integration, multi-source complementary" mode, thereby achieving a high degree of coordination and interoperability of source-network-load-storage (Zhang et al., 2020). The SES further combines the multi-energy coupling system with information technology, computer technology, etc., by changing the energy consumption mode, controlling the size and direction of the energy flow, thus realizing intelligent multi-energy flow collaborative management and energy conversion, and meeting the optimal allocation of resources under the balance of supplier and demander.

The early research on multi-energy flow mainly focused on steady-state energy flow and dynamic energy flow. The steady-state energy flow can reflect the system's stable operation for a period, laying the foundation for research into collaborative planning and optimized operation of SESs (Chen et al., 2020). However, it ignores the problem of different time scales between different energy sources, and fails to achieve the matching and balance between "fast" transmission of electricity and "slow" transmission of heat and gas. Therefore, it is hard to analyze the system's dynamic response caused by various small disturbances in real life, limiting the real-time control and scheduling of the system operation. In order to break through this limitation, many scholars have turned their attention to the study of dynamic energy flow. Dynamic analysis can better reflect the dynamic process of the system, realize real-time control and scheduling of system operation status, and significantly improve energy utilization efficiency (Yao et al., 2020). However, there are a lot of partial differential equations that are difficult to solve directly in the dynamic energy flow model. At present, some scholars use the method of Wendroff difference form (Fang et al., 2017) and Taylor's expansion (Pan et al., 2019) respectively to deal with it to reduce the difficulty of solving it. Therefore, in the study of dynamic energy flow in SESs, it is of great significance to explore more efficient and accurate differential solution methods and seek innovation and improvement in differential equations and nonlinear equation linearization methods for dynamic model solving.

Later, with the continuous deepening of energy flow research, probabilistic energy flow and optimal energy flow have gradually become research hotspots (Khorsand and Seifi, 2018). Probabilistic energy flow can better consider the uncertain factors in SESs (such as the volatility of RE and the randomness of the load) to provide a reference for the preparation of emergency measures or backup plans for the system (Dong et al., 2019). However, the current probability energy flow is usually described by some traditional specific distribution functions, making the model relatively rough and inaccurate. Therefore, exploring more advanced mathematical methods or AI technology is significant to probabilistic energy flow modeling. Optimal energy flow aims to explore the characteristics of coordination and coupling between multiple energy flows, which can help the system find a balance point between economic optimization, the lowest emissions, the least pollution, and other goals (Ceseña and Mancarella, 2018). However, many constraints need to be considered in calculating optimal multi-energy flow, which makes its solution more difficult. Therefore, exploring efficient intelligent algorithms for solving problems is of great significance to the research of optimal energy flow. Table 5 describes the different energy flow research methods and their characteristics and summarizes the existing research work.

Table 4
Engineering projects with typical DR application.

Countries	Project	Specific measures
USA (Huang, 2010; Huang et al., 2010)	Day-ahead real-time price	Nyagana Power: Customers receive hourly prices for up to 24 h a day in advance to schedule future electricity usage.
	Mandatory real-time price	New York, New Jersey, Pennsylvania power companies are implementing real-time prices for large industrial users, and the scope is expanding.
	Day-ahead DR of NYISO	Day-ahead market determines the electricity price at the time of clearing, users bid with reduced load and price.
	EDR project of PJM	The load participating in emergency response will be compensated according to the nodal marginal price.
	ASM project of PJM	ASM project is divided into 30 min operation reserve day-ahead market and 10 min rotating reserve real-time market. Data measurement is required for a period of less than 1 min load.
	Opower energy management company	It analyzes users' energy consumption behaviors and provides energy consumption reduction schemes based on big data and cloud computing, thus realizing the integrated application of energy consumption and internet.
France (Zhang, 2019)	Dividing electricity price according to user's reported installed capacity	Green electricity price: it is used for industrial users above 250 kVA and is set according to the user's preset power consumption. Yellow electricity price: it is used for commercial users with 36–250 kVA. Which is set according to the time of power consumption for users. Blue electricity price: it is used for 3–36 kVA residential users, which has the characteristics of high utilization hours and low electricity price.
China (Bahrami and Sheikhi, 2015; Xu, 2019)	TOU price in Shandong province	The power consumption time is divided into peak and valley periods to improve the price difference.
	Cut prices in Zhejiang province	Reduce the price of industry and commerce power consumption, distinguish different voltage level user, and implement peak valley price.
	Load shifting and power limitation in Guangdong	By customizing energy-saving solutions and DR strategies, enterprises can actively reduce power load in peak hours.
	IES service platform in Jiangsu	The "Internet +" and big data mining technologies will be used to predict the load capacity of electricity, gas and heat as well as RE generation, so as to fully mobilize users to participate in peak adjustment and DR of the power grid in the park.
Demark (Lund et al., 2005)	Electricity-gas correlation system	Extensively develop CHP, heat pump and other technologies, establish district heating systems, and integrate industrial loads around thermal power plants to adjust peaks and improve energy utilization efficiency.
Britain (K Peng et al., 2017)	User interaction platform of IES	Realize the step utilization of various resources, collect users' energy information through terminal monitoring equipment, establish a DSM platform for multi energy to control the load state and alleviate the ES.
	TOU price Real-time price	The UK powerhouse will collect day's power plant operation data and power load forecast data every afternoon to determine the next day's generation plan and TOU price.
Germany (Chenhui et al., 2015)	E-telligence Demonstration Project	It involves a variety of energy conversion equipment, establishes TOU and real-time pricing mechanisms, and reduces the volatility of new energy output through centralized management of virtual power plants.
Japan (Zhang, 2019)	JSCA engineering project	Develop integrated community energy systems including electricity, gas, heat, and renewable energy, and IES with transportation, water supply, information and healthcare systems.

In addition, energy conversion technology can realize the mutual conversion of multiple energies and provide a path for the cascade utilization of multiple energy resources (Zhang, B. et al., 2020). Current research includes the study of the energy converter itself and the EH that integrates multiple energy converters. The former mainly studies the characteristics of the energy converter itself, such as structural design, control strategies, performance evaluation indicators, etc. (Geidl et al., 2006). Table 6 lists the individual energy conversion elements and their functional characteristics and summarizes the existing research literature. The latter is to explore the EH's overall planning, operation, and regulation characteristics, aiming to realize the comprehensive utilization of multiple energy resources (Sheikhi et al., 2015). In addition, EH has the characteristics of small scale and flexible operation, which can provide a peak shaving function for the power grid. Based on this, Mohammadi et al. (2018) comprehensively reviewed the different applications of EH in various energy consumption sectors from the aspect of realization of energy system models and towards sustainable MESs, aiming to evaluate the potential role of EH in solving the main challenges of these consumption sectors. Mohammadi et al. (2017) discussed the weaknesses, advantages, and challenges of the EH model from the aspect of main structure of the EH, aiming to identify these challenges to introduce new options for the EH model and explore its potential as a comprehensive model for future sustainable energy systems. Sadeghi et al. (2019), based on the operation and expansion planning of the energy system in the context of EH, explored in detail the problems, difficulties, importance, and necessity of EH in the power industry application field. Geng et al. (2020) explored the capacity optimization design and operation of an EH that powers the island load, and studied the shared energy between multiple EHs, aiming to improve the reliability of the EH while reducing operating costs.

Multi-energy flow collaborative management and energy conversion technology are the keys to the stable operation of SESs. The analysis found that the technology also has some shortcomings in the development process. For one thing, the research of multi-energy flow

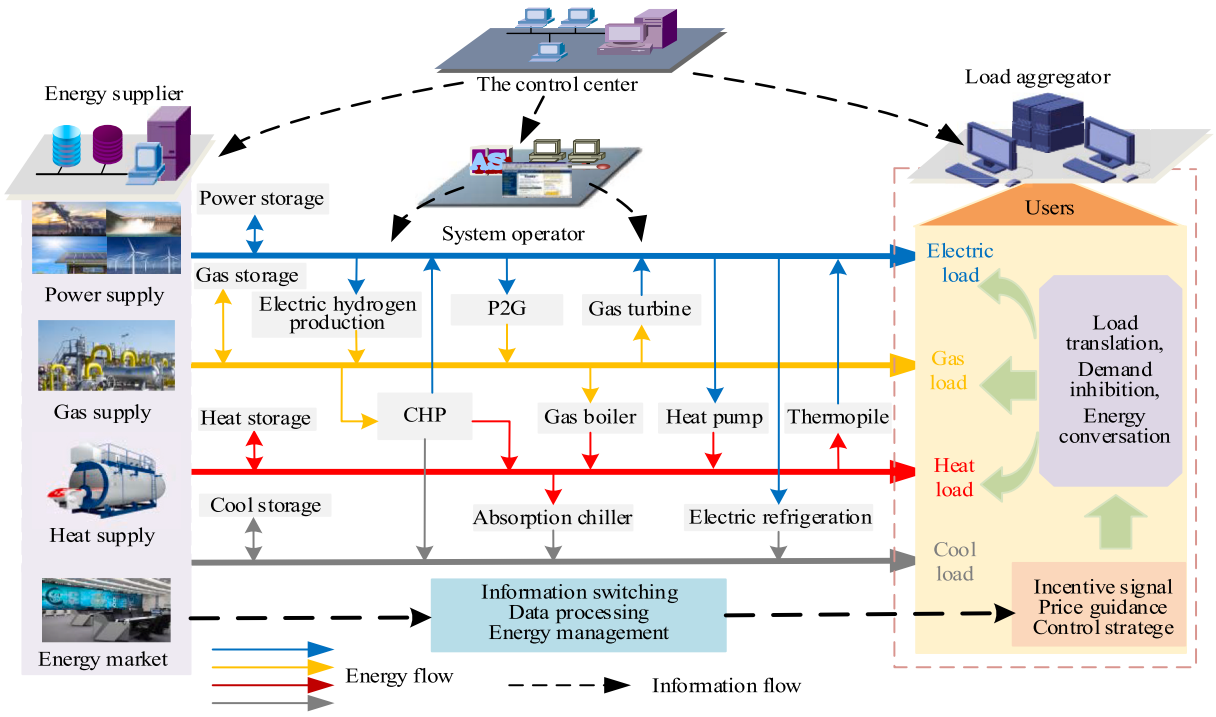


Fig. 6. Multi-energy flow collaborative management and energy conversion.

collaborative management mainly focuses on establishing different energy flow models. Meanwhile, there are relatively few studies on precise control of the direction of mutually coupled multi-energy flows. Moreover, the constraining characteristics of different energies that accompany the energy flow process will impact the coordinated management of multi-energy flow. For another, the current research on energy conversion focuses on the input and output characteristics of the conversion device, while the dynamic characteristics of the coupling element itself and the energy hysteresis characteristics are seldom studied. Therefore, in the future, this technology should deeply integrate intelligent technology to realize real-time monitoring and management of energy conversion in the system and evaluate the impact of small changes in energy flow on SES in real-time, thus quickly taking emergency response measures.

3.4. Advanced energy storage technologies

As shown in Fig. 7 (a), advanced ES technology combines multidisciplinary cross-discipline research, covering thermodynamics, kinetics, microstructure, fluid mechanics, chemical reactions, physical evolution, etc., which effectively realizing the efficient storage of various energy (Liu and Du, 2020). It helps to address the uncertainty of distributed generation and load in energy usage and improves the energy supply reliability and flexibility (Nazari-Heris et al., 2020). ES technology can be divided into physical ES (Afif et al., 2019) and chemical ES (Yoo et al., 2014), whose detailed classification is shown in Fig. 7 (b). Physical ES include pumped storage, compressed air ES, flywheel ES, superconducting magnetic ES. Chemical ES include lithium-ion batteries, sodium-ion batteries, flow batteries, sodium-sulfur batteries, lead-acid batteries, supercapacitors, and other new battery ES. Table 7 describes the performance characteristics and applications of different ES technologies from the perspectives of discharge depth, discharge efficiency, action frequency and discharge duration.

Pumped storage is one of the most important large-scale ES technologies globally, whose current installed capacity in the world exceeds 127 GW (Joseph and Chelliah, 2017). Pumped storage realizes the mutual conversion between electricity, potential energy and mechanical energy through several interconnected reservoirs located at different heights, thereby effectively solving the unreliability of RE power generation and power grid's peak/frequency regulation problems (Ahmadi et al., 2020). The installed capacity of China's pumped storage power plants has ranked first in the world, which was 30 GW at the end of 2018, accounting for 1.6% of the total installed capacity of power generation. It is predicted that China's installed wind pumped storage capacity will reach 130 GW by 2030. In the context of the vigorous development of SESs, pumped storage has also received new research. For example, a small and isolated sustainable power generation system comprising photovoltaics with pumped storage and an open well was studied, which effectively solved the difficulty in obtaining electricity in remote areas (Pali and Vadhera, 2020). The closed-loop variable-speed pumped storage hydropower plant and the secondary regulation reserve scheduling model were explored, which effectively solved the energy loss problem of electronic frequency converters (Chazarra et al., 2017). However, there are also some problems with pumped storage: 1) The construction of pumped storage power plants has geographical restrictions, as the upper and lower reservoirs must exist within a relatively short distance and have a high height difference. 2) Under the condition of little height difference, the energy density that a pumped storage power station can achieve is minimal. 3) Pumped storage power plants have low economy, high investment costs, and long payback periods, usually requiring a payback period of more than 30 years.

Compressed air ES stores energy in high-pressure air and drives turbines to generate electricity by burning high-pressure air and other gas fuels (Tong et al., 2021). Its scale is second only to pumped storage, and it has the advantages of longer charge/discharge times, so it has also been studied and applied on a large scale. For example, the operation mode and safety issues of different compressed air ES system types were explored, which effectively evaluated performance and application prospects (Olabi et al., 2020). A new hybrid wind turbine system formed by a continuously variable transmission connection of the turbine drive shaft with an air compressor was explored, thus effectively solving the problem of smooth electricity output under fluctuating wind speed (Krupke et al., 2016). A new heat storage strategy in adiabatic compressed air ES systems was explored, that is, the use of multi-phase

Table 5
Research characteristics of multi-energy flow in smart energy system.

Type	Characteristics	Ref.	Main research contents/methods
Steady-state energy flow	It can reflect the stable system operation status for a period, but ignores the problem of different time scales between various energy flows.	Zhang et al. (2020)	Uses a distributed sequence to calculate the energy flow distribution.
		Chen et al. (2020)	Proposed a fast method for decoupling multi-energy flow.
		Massrur et al. (2018a)	Uses a non-iterative method to solve the energy flow of large-scale systems.
Dynamic energy flow	It can realize real-time control and scheduling of the system's running state, but a lot of partial differential equations make it difficult to solve.	Yao et al. (2020)	Established a thermoelectric integration dynamic energy flow model based on the finite difference method.
		Fan et al. (2020)	Proposed a dynamic energy flow model considering the gas inertia.
		He et al. (2020)	Uses a multi-time scale hybrid simulation algorithm to solve the dynamic heat flow model with migration characteristics.
Probabilistic energy flow	It can better consider the uncertain factors in the multi-energy system, but the traditional distribution function model usually used is not accurate enough.	Dong et al. (2019)	Proposed a probabilistic mixed energy flow assessment method based on Dempster-Shafer evidence theory.
		Massrur et al. (2018b)	Proposed a 2 m + 1 point estimation strategy for analyzing probabilistic energy flow.
		Sun et al. (2017)	Proposed a multi-period comprehensive probabilistic power flow model.
Optimal energy flow	It can help the system find an optimal operating state, but the existence of a lot of complex constraints makes it difficult to solve.	(Ceseña and Mancarella, 2018)	Uses two-stage iterative modeling to explore the optimal energy flow.
		Kou et al. (2019)	Proposed a multi-period optimal energy flow model based on ADMM.
		Qu et al. (2018)	Proposes a decentralized optimal energy flow in a carbon trading market to solve the difficulty of information collection in sub-regions.

materials, which effectively solved the problem of heat loss in the system (Li et al., 2021). In terms of application, the typical representatives of compressed air ES power stations are Huntorf in Germany and McIntosh in the USA. In addition, the UK has established a liquid air ES with capacity of 5 MW. China has built the world's first 1.5 MW supercritical compressed air ES system demonstration platform and completed the design of a 10 MW supercritical compressed air ES system. Various new types of compressed air ES technologies, such as adiabatic, liquid, and supercritical, have been proposed to eliminate geographical and resource constraints. However, their efficiency is less than 70%. In the future, the development of compressed air ES technology that does not

Table 6
Functional characteristics of energy conversion elements.

Energy converter	Characteristics	Ref.	Main research content
Gas generator	Compared with coal-fired generators, it has high thermal efficiency, low cost, high reliability, and low environmental impact.	Palmieri et al. (2021)	Design of Innovative load controller for Heavy-Duty gas turbine generator.
		Bonfiglio et al. (2018)	Gas turbine control method based on sliding mode model.
		Park et al. (2013)	Heat transfer and temperature distribution of micro gas turbine system.
Gas boiler	Enhancing the coupling relationship between the natural gas system and the thermal system.	Liu et al. (2020)	Emissions and thermal efficiency for premixed burners in a condensing gas boiler
		Wang et al. (2020)	Optimal scheduling of energy consumption in a smart home with a hybrid gas boiler.
Combined heat and power (CHP)	In the process of gas-fired power generation, CHP uses the heat recovery system to capture waste heat and output heat energy to improve energy efficiency.	Zhu et al. (2021)	Development status of CHP technology.
		Lin et al. (2019)	Analysis of Structural Characteristics of Cogeneration System.
		Li et al. (2015)	Optimal scheduling of IES considering CHP operation characteristics
Power to gas (P2G)	Hydrogen produced by electricity further produces methane through the methanation reaction.	Clegg and Mancarella (2015)	Explore the integrated model that simulates the P2G system.
		Zhang et al. (2022)	The multifunctional characteristics and application potential of P2G technology.
		(Quarton and Samsatli, 2018)	P2G technology considers real projects, economic evaluation, and system modeling.
		Thema et al. (2019)	Explore P2G projects focusing on hydrogen production or renewable alternative natural gas.
Electric hydrogen	Hydrogen production by water electrolysis, a large amount of hydrogen storage, promote the use of clean energy.	(d'Amore-Domenech et al., 2020)	Multicriteria analysis of seawater electrolysis technologies for hydrogen production.
		Wang et al. (2014)	Kinetic and thermodynamic analysis of hydrogen production technology by electrolysis of water.
Electric boiler		Lei et al. (2021)	An active cut-off air regulation model

(continued on next page)

Table 6 (*continued*)

Energy converter	Characteristics	Ref.	Main research content
	Directly convert electrical energy into heat.		based on heating of regenerative electric boiler is proposed.
		(Li et al., 2021)	Exploring the influence of regenerative electric boilers on the integration of wind power.
Heat pump	Transfer the heat energy of the low heat source to the high heat source.	Fan et al. (2021)	Explored the application of heat pumps in solar technology.
		Xie et al. (2021)	The regional applicability of heat pumps is analyzed based on technology and economy.
Absorption chiller	By absorbing high temperature heat to drive the chiller to work, the coupling of cold and heat energy and the cascade utilization of energy are realized.	Rad and Davoodi (2021)	A combined drive single effect absorption chiller is analyzed.
		Kühn et al. (2020)	Experimental investigation of ionic liquids as substitute for lithium bromide in absorption chillers.
		Zhao et al. (2020)	extends the developed heat current method to single effect lithium bromide absorption chiller.

rely on fossil fuels and large air storage chambers should be achieved mainly by improving the technical performance of key components, optimizing system integration and control technology.

Flywheel ES is currently one of the ES devices with the longest energy storage life, and it is especially suitable for small-scale and short-period applications (Lee et al., 2010). In recent years, flywheel ES has entered a period of rapid development, forming an upsurge in the research and development of flywheel ES single technology and integrated application technology. For example, the applicability evaluation of flywheel ES technology was explored in the operation of large-scale electric vehicles, thereby providing technical support for solving the problem of a lot of backup energy losses (Thormann et al., 2021). A flywheel storage system control strategy based on permanent magnet-assisted synchronous reluctance motor heating was designed, which effectively solved the problems of memory system bias and dead zone resistance (Zarbil et al., 2021). An optimal control strategy for the output stability of a flywheel ES based on energy function was explored, which effectively solved the problem of isolation between wind power output and voltage fluctuations (Ghosh and Kamalasadan, 2016). In addition, flywheel ES has been applied in many fields such as electric vehicles, uninterruptible power supplies, orbital satellite ES. For example, Amber Kinetics has created an ES project with a power of 20 MW and capacity of 80 MWh. The United States Beacon Power established a flywheel ES substation with power of 20 MW and capacity of 5 MWh. Australia has equipped wind turbines with a 500kVA flywheel ES system. Flywheel ES currently faces some problems: 1) The energy density is low and self-discharge rate is high, the motor rotor manufacturing cost is high, and friction loss is considerable; 2) Most of the related research on flywheel power regulators stay on UPS application, which is not ideal for grid stability control and power quality improvement.

Superconducting magnetic ES stores electromagnetic energy in a superconducting energy storage coil, which has a fast response, high conversion efficiency, and high efficiency dynamic power transfer (Mukherjee and Rao, 2019). It can effectively improve the reliability of the power supply and the dynamic stability of the power grid, so it has also become a research hotspot in academia. At present, scholars have carried out research from the instantaneous support of superconducting magnetic energy storage under short-term disturbances in the power grid (Kouache et al., 2020), the performance problems of different solenoids (Pan et al., 2016), and the regulatory framework and economics of energy storage systems (Colmenar-Santos et al., 2018), respectively. In terms of applications, the 21 MW superconducting ES power station designed by the United States, and the 1 MJ/0.5 MVA high temperature superconducting ES substation implemented by China have typical reference significance. It is worth noting that the main shortcomings of superconducting ES: 1) The price of superconducting materials is still higher, and the critical temperature needs to be increased; 2) The low-temperature refrigeration system applied by superconducting technology is complex, and the maintenance-free life of the refrigerator is shorter, which limits its industrialization and application. 3) The low-temperature and high-voltage insulation technology of superconducting equipment, real-time detection technology, integration technology, and matching and coordinated operation with conventional systems also need further research.

Chemical ES has received widespread attention due to its simplicity and high efficiency, and it has become the main development direction in many ES applications (Miller et al., 2021). Compared with other forms of electric ES, the chemical ES technology represented by batteries has the largest variety, the fastest development, and desirable economic viability (Mao et al., 2015). Further, its different characteristics can meet various needs in the power system. As the core technology and bottleneck technology of electric vehicles (Mao et al., 2019), chemical ES is significant to the massive adoption of new electric vehicles and energy saving and emission reduction in our country. To this end, a variety of new battery ES technologies, including battery life (Alihosseini and Shafaee, 2021), battery recycling (Chandran et al., 2021), battery material design (Ding et al., 2021), battery energy management strategy (Rezzak and Boudjerda, 2020), battery capacity (Qian et al., 2021), etc., have been researched and made some progress. In terms of applications, it currently includes demonstration projects such as the lead-carbon battery ES project with a capacity of 3MW/1.4 MWh built in the United States, and the all-vanadium redox flow battery ES station with a capacity of 200MW/800 MWh built in China. Some problems chemical ES faces during the rapid development: 1) How to break through the key technical indicators of battery products? Such as improving battery cycle life, specific energy, specific power, high and low temperature resistance, low current, and deep discharge performance; 2) How to steadily improve the cost-effectiveness of battery products? Such as developing low-cost, high specific surface area activated carbon materials, thus reducing battery manufacturing costs and greatly extending cycle life.

3.5. *Information exchange technology based on high-speed network*

The term "information and communication technology" refers to the process of transmitting, exchanging, and storing data (Xiao et al., 2020). It can integrate data on energy production, grid operation, user demand, and the energy market, among other things, and is a critical component of realizing the management and dispatch of SESs (Wang and Perera, 2020). Fig. 8 illustrates the effect of information exchange technology on the connectivity between the energy system's operational, data, and physical layers. We will primarily study and examine the three information exchange technologies used in the system: energy routers, cloud and 5G technology, and blockchain.

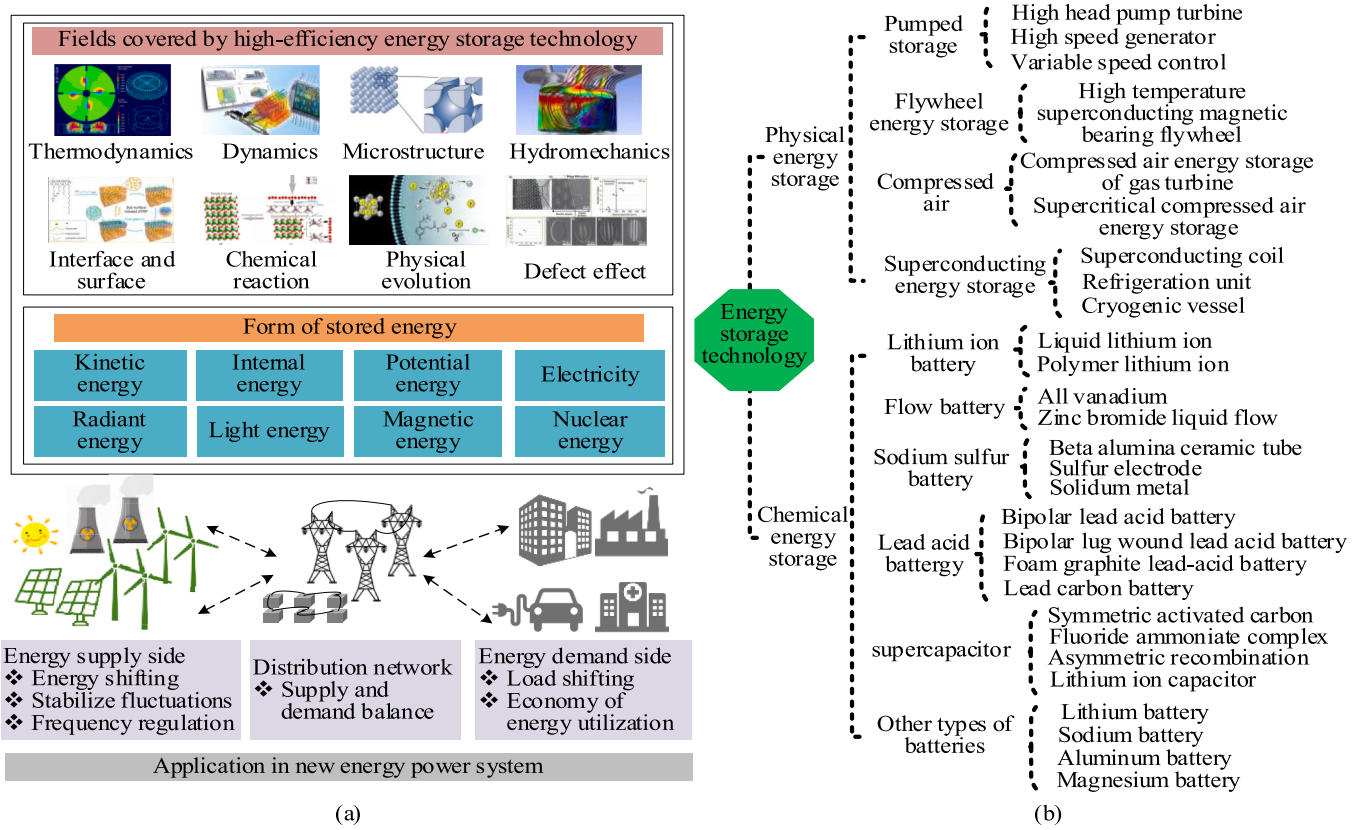


Fig. 7. Advanced energy storage technologies: (a) Classification of energy storage technology, (b) Application of energy storage technology in new energy power system.

Table 7
Energy storage technology performance characteristics and applications.

Type	Discharge depth	Discharge efficiency	Action frequency	Discharge duration	Applications
Pumped storage	Deep	75%–82%	Frequently	Hours	System backup
Compressed air ES	Deep	40%–55% (non-adiabatic system) 60%–70% (adiabatic system)	Frequently	Hours	Balance load Improve system transient stability
Flywheel ES	Shallow	>90%	Occasionally	Milliseconds Seconds Minutes	Improve power quality Uninterruptible power supply
Superconducting magnetic ES	Shallow	>90%	Occasionally	Milliseconds Seconds	Dynamic voltage stability Power reliability
Chemical ES	Deep	>90%	Occasionally Frequently	Seconds Minutes Hours	Improve transmission capacity

1) Energy router is a highly integrated communication system powered by electronic information technology (Alqahtani and Hu, 2020), which can receive, process, and share information flow in real time. Based on the current research and application of energy routers, different scholars have solved different problems. For example, a hierarchical optimization strategy based on energy routers was proposed to solve privacy protection problems (Guo et al., 2019). A quantity buffer was proposed to solve the problem of network real-time control loss (Xia et al., 2021). An improved energy routing algorithm was explored to solve standardization and interoperability communication problems (Hussain et al., 2019).
2) Cloud technology and 5G networks have made SESs gradually establish a distributed shared database (Hui et al., 2020). Beginning with the information exchange service, researchers investigate how information is expressed and system interoperability during the data exchange process to increase the security and transparency of information sharing (Priyanka et al., 2021). Based on this, a two-tier effective energy management system based on energy sharing clouds was proposed to solve the ever-changing energy demands of cloud users (Li et al., 2021). An evaluation framework using edge computing and 5G technology was proposed to realize the optimization management problem of energy cloud systems (Jararweh and Computing, 2020). A cloud-based multi-objective energy management strategy and training method were proposed to solve the problem of energy security (Li et al., 2021).
3) Blockchain technology makes use of distributed ledgers to facilitate the exchange of transparent, reliable, and traceable data between network nodes (Noor et al., 2018; Wang and Su, 2020), effectively connecting power grids, new energy companies, regulatory agencies, financial institutions, users, and other entities to achieve data penetration and sharing across the industry chain (Ahl et al., 2019). The applicability of blockchain-based decentralization technology in the energy field was discussed, and the potential of blockchain energy and its different utilization possibilities was explored from the perspective of energy transition research (Teufel et al., 2019). An optimization strategy for IoT sensing sampling using blockchain and Tangle technology was proposed to minimize the information period experienced by end users, thereby effectively solving the slowness of

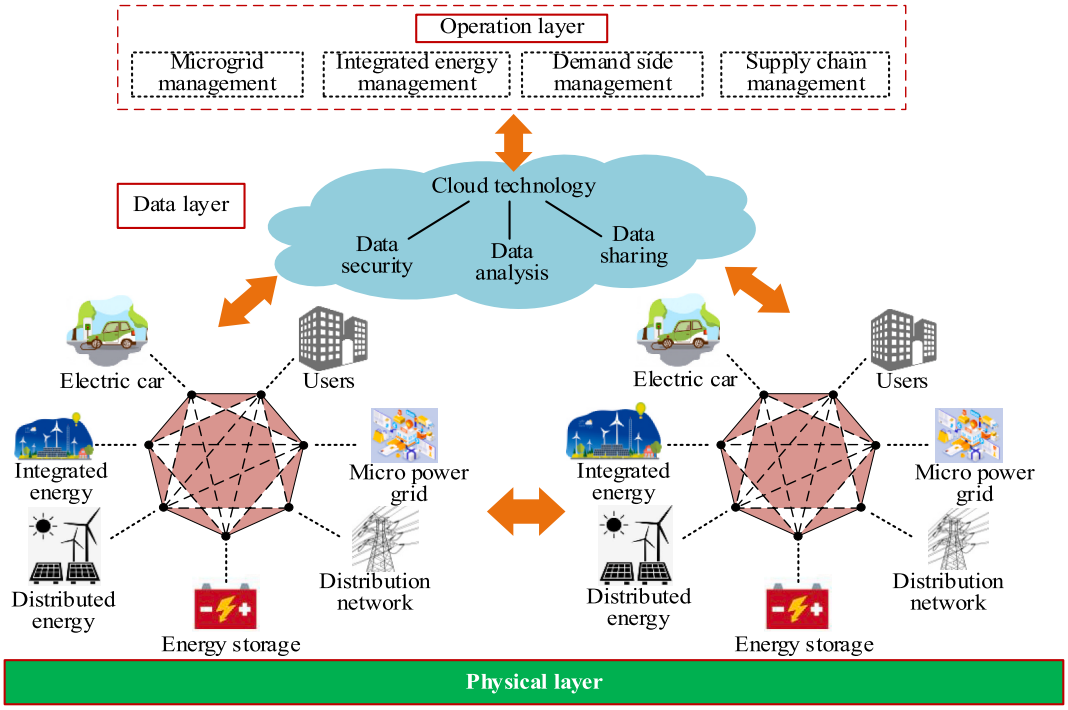


Fig. 8. Information exchange technology of SESs.

the technology in consensus-based information verification (Rovira-Sugranes and Razi, 2019). A network microgrid scheme that uses blockchain technology was explored to optimize the financial and physical operation of the SES, aiming to use the powerful potential of blockchain in distributed data storage to improve the microgrid management of energy transactions (Li et al., 2019). Considering the heterogeneity of demand flexibility associated with different types of users, a safe and efficient SES through smart contracts and blockchain technology was created to realize flexible interaction between participants, and the flattening of grid power procurement plans (Li et al., 2019).

In summary, the current research on advanced information and communication technologies in SESs mainly focuses on academic research, and large-scale applications in real life are not mature enough. Hence, SES has not been implemented on a large scale in actual society. For this reason, both the mature application of information and communication technology in the SES and the information exchange between the various energy-using sectors can effectively promote the actual large-scale SES implementation. This will be the only way for the development of smart cities in the future.

3.6. *Digital-energy integrated market and service mechanism*

As a result of the digital revolution, building a fully integrated digital-energy market and service mechanism is critical for advancing the establishment of smart grids and smart cities. With the reform of the power system and the gradual openness of the multi-energy market

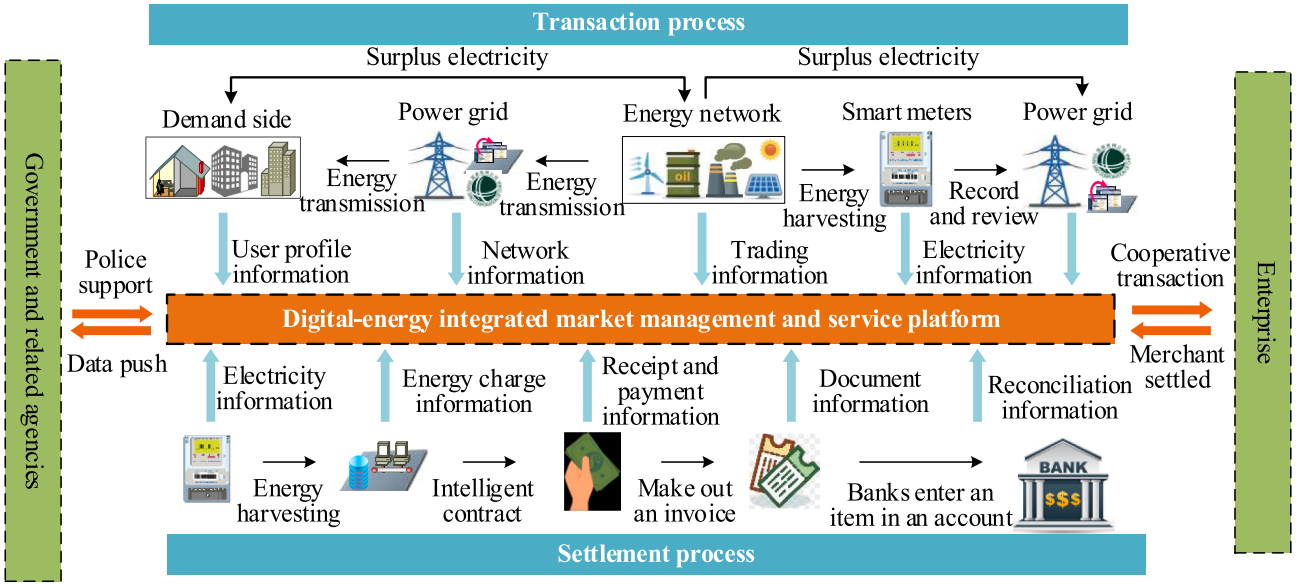


Fig. 9. Digital-energy integrated market and service platform.

(Aghaei et al., 2009), the traditional model of selling electricity is evolving into full service with different energy kinds and various price models (Naughton et al., 2020). Fig. 9 shows the digital-energy integrated market and service platform. Under the trend of increasingly diversified market entities, digital platforms have increasingly proposed business models to improve economic organization and achieve higher efficiency by coordinating supply and demand under incomplete information.

For this reason, domestic and foreign scholars have carried out research and application on it from different angles. For example, to explore the importance of market size, digital readiness, and regulatory quality in the market (Duch-Brown and Rossetti, 2020), and (Hiteva and Foxon, 2021) respectively surveyed 217 digital energy platforms of the EU market and the socio-economic value created by digital energy services for users and energy systems. In order to explore the structural characteristics of the energy market and its energy management issues, some scholars have researched the establishment of an evaluation model for controlling multiple energy carriers in the energy market (Köse and Sauer, 2021), the assessment of the potential of multiple energy communities to bring flexibility to the energy market (Good and Mancarella, 2017), the exploration of energy management methods for buildings that can be connected to different power markets under limited information (Lesage-Landry et al., 2019), the establishment of a two-level market model using intelligent perception (Liu et al., 2015) and the establishment of a digital energy management platform based on the Internet of Things (Khani et al., 2015). It aims to establish an accurate multi-energy market structure model and provide technical support for the coordinated operation of the digital energy market.

In addition, the digital energy market trading platform serves as a bridge for exchanges and transactions between various entities in the market. The privacy, security, and transparency of various types of information in the transaction process are crucial. It is also the key to whether the entire transaction process can proceed normally. Therefore, to explore the characteristics of information transmission in the digital energy market transaction process, a secure energy transaction method using homomorphic encryption technology to solve the privacy leakage problem in energy transactions was explored (Yi et al., 2021). The blockchain consensus mechanism and smart contracts were used (Chen et al., 2018) to solve the problem of information asymmetry in the transaction system. An efficient and collaborative information chain storage technology was used in (Miglani et al., 2020) to improve user participation and satisfaction. These studies have provided theoretical support for the digital energy market trading platform to implement point-to-point transactions between energy entities, improve the security of energy transactions, prevent the leakage of user information, and improve the overall service quality of the trading platform. However, this technology faces a similar problem to that of information and communication technology, that is, what is the degree and breadth of the technology's application in society in the short term. In addition, the market service mechanism involves a variety of subjects, and the implementation of a mechanism should consider the operating characteristics and participation willingness of multiple entities. Therefore, operating an innovative market mechanism under the condition of weighing multiple stakeholders and realizing the active participation of the masses will be an urgent problem to be overcome in the development of this technology.

4. Challenges for smart energy system in the context of carbon neutrality

Energy production, distribution, and consumption have steadily attained liberalization, diversification, and intelligence due to the rapid development of the national economy and continual advances in science and technology. The increasing expectations placed on the energy service business by society necessitate ongoing innovation and advances in a variety of SES technologies. As a result, the primary problems confronting critical SES technologies in the context of carbon neutrality are as follows:

1) **Intelligent prediction technology with multiple uncertainties:** The current prediction technology mainly focuses on forecasting a single entity. For example, a photovoltaic power generation prediction technology is applied in a system containing multiple distributed energy sources. However, this kind of prediction technology only analyzes the law of photovoltaic power generation. It is difficult to simultaneously predict other uncertain factors in the system, such as wind energy volatility and user energy randomness. In addition, the current prediction technology is immature. For example, the prediction of some extreme accidents is imprecise, and the prediction and analysis of the damage that imposes on the system after such an accident is not specific enough. Therefore, it is necessary to further study an efficient prediction technology that can perform unified real-time online analysis of the multiple uncertainties of multiple entities in the SES.
2) **Intelligent emergency response to extreme disasters:** The major power outage in Texas in February 2021 was mainly due to the fact that the local power generation equipment and devices in Texas did not take into account the severe cold weather, which made them unable to operate in the face of extreme weather (Fisher Allison, 2021). In addition, all energy sources in Texas, including RE, coal, natural gas, nuclear power generation, etc., are affected by the local blizzard, causing the energy system to be frozen and the power plant equipment unable to operate (Ferman, 2021). Such a blackout has attracted worldwide attention, which has triggered extensive discussions and in-depth thinking in academia and industry. For this reason, the development of an efficient emergency response system to provide stable energy supply and demand in extreme disaster situations is an essential issue of SES's key technology.
3) **Power supply problems in remote mountainous areas:** SESs closely rely on efficient intelligent technology, thus it is mainly established in areas with more advanced information technology. However, in areas where conditions are relatively backward in all aspects, especially in remote mountainous areas where information technology has not yet been popularized, it will be challenging to build SES. Therefore, guaranteeing the energy needs of people in remote areas, building an intelligent energy supply system, and realizing intelligent energy use technology that is not restricted by geographic location will be a major challenge for SES construction in the future.
4) **How to adapt infrastructure to extreme climates:** A stable and reliable infrastructure is essential to develop of SESs, and it can support the efficient implementation of various key technologies in the system. At present, the main difficulty facing the infrastructure construction of the power supply system is maintaining stability in the face of major natural disasters without affecting the reliability of the energy supply. Therefore, comprehensively studying the construction of a solid infrastructure in terms of materials, structure, installation location, etc., thus improving the toughness of SES, will be an urgent problem that needs to be overcome in the future development of SES.
5) **How to solve the problem of massive waste of energy:** The existing social energy waste is significant, with most of it focused on commercial building electricity usage, especially the central air-conditioning system. Therefore, on the one hand, how to strengthen citizens' energy-saving concepts, build an effective penalty mechanism, engage residents in energy-saving actions, increase energy usage efficiency, and reduce energy waste is a significant problem in the building of SESs. On the other hand, comprehensively exploring more efficient intelligent equipment control technology from the aspects of economy, environmental

protection, and energy consumption comfort, thus implementing precise energy consumption control, is also the key to the development of SES.

6) **Multi-energy market joint planning problem:** Distinct energy markets operate on a different time frame and follow a different set of clearing criteria. This prevented the enthusiasm of multiple entities in the SES from participating in the market to a certain extent. The complex market operation mechanism also impacted the implementation of DR. Therefore, how to organize numerous energy markets collaboratively to achieve concurrent operation, rationalize healthy market competition, and simplify the complicated mechanics of multi-energy markets will also be important issues facing the development of SESs.
7) **New energy comprehensive development and utilization technology:** There are various forms of energy on the earth, and the new energy that is currently being developed and put into the application is only the tip of the iceberg, and the development and utilization of new energy is far from enough. For this reason, how to gather the scattered and diverse energy in society and how to protect the natural environment while further mining and using RE on the earth is crucial to the growth of SESs.
8) **The challenge of large-scale explosive diseases to the construction of SESs:** Large-scale explosive diseases, such as the COVID-19 pandemic, have severely affected human life, production, and work methods. They have also largely impacted the energy production and consumption patterns of the entire society (Zhang et al., 2021). As a result, it is crucial to investigate more efficient and advanced intelligent technologies, so that the SES system can maintain a relatively stable operation in the face of major epidemic disasters or major changes in human energy consumption characteristics. These efforts can ensure that people's material, cultural, and spiritual requirements are met during a disease outbreak.
9) Large-scale and normal use of carbon capture and carbon recovery technologies: The development of SES is one of the critical means to promote the realization of carbon neutrality. Therefore, the rational and efficient use of carbon capture recovery technologies in the system are crucial. However, the current carbon capture and recovery technology is still in their infancy, and it mainly has problems such as difficulty in application, high cost, and low efficiency. Therefore, in the context of the rapid development of intelligent technology, overcoming these obstacles and realizing the large-scale and normal application of high-efficiency carbon capture and recovery technology is a major challenge for the construction and development of SESs.
10) **Emergency handling of abnormal behaviors and sensitive access in the information exchange process:** Information conversion and transmission is the bridge to realize SES operation, but it also faces security risks. To this end, while protecting the security and privacy of information and data, exploring a more efficient intelligent technology to realize real-time monitoring of the information exchange process is important. In addition, how to take timely countermeasures to deal with abnormal behaviors and sensitive access in the process of information exchange, and to resolve the contradiction between open information sharing and data privacy protection, will be a major challenge for the large-scale implementation of SESs in the future.

5. Opportunities for smart energy system in the context of carbon neutrality

With the active advancement of global carbon neutrality efforts and the development of smart cities, as well as the continued development of intelligent information technologies such as AI, big data, cloud computing, and the Internet of Things, the construction and development of SESs face new opportunities:

1) Government agencies strongly promote the development of SESs in the context of energy conservation and emission reduction. The primary objective of the SESs is to achieve energy transformation, increase energy efficiency, and facilitate the attainment of carbon peak and carbon neutral targets. SES development must coordinate with national development objectives and institutional changes, and the pace of important technology deployment must equal that of reforms. As a result, a significant number of specialists and academics in academia and industry have conducted extensive research on it with the strong assistance of government authorities. This will enable a qualitative leap forward in critical SES technologies in the future, while also promoting the widespread and efficient usage of technology in the actual world.
2) Promote the development of high-density, low-cost, and long-life ES device materials. The application of large-scale and high-proportion RE has resulted in the continual development of high-efficiency and advanced ES technologies. New subject areas are continually being discovered because of scholars' in-depth examination of multidisciplinary studies. This revitalizes the efficient ES business and creates new potential for the future realization of additional large-scale and steady ES forms.
3) Promote the construction of DR precision service projects using user portraits as a starting point. With the continual advancement of intelligent technology, the DR technology embedded in the intelligent energy system faces new opportunities. Utilize massive data to automatically extract and understand the characteristics of user samples, and to conduct an in-depth analysis of users' energy needs and random behaviors, to facilitate the construction of realistic disaster recovery service projects based on user portraits.
4) Promote the establishment of personalized and innovative SES services, and actively develop breakthrough technologies that consider the needs of society and people's livelihoods. The adaptability of the system's resources and the active participation of various stakeholders have facilitated the transition from traditional material energy consumption to integrated energy, information, and service consumption. Additionally, the advancement of intelligent information technology will contribute to the expansion of the future energy integrated service business model and system, and the enhancement of the digital energy integrated market and service management mechanism.

6. Conclusions and future works

6.1. Conclusions

SESs have been explored and developed regularly in recent years to promote sustainable development of smart cities. The SES offers flexible multi-energy control and management under various unpredictable or harsh settings, while also delivering more diverse sources and uses of energy. To that purpose, this article explores the research and application characteristics of six key hotpot technologies in SESs and analyzes their existing inadequacies. Then, in light of global attempts to attain carbon neutrality, this article examines the ten significant obstacles and four major opportunities confronting SESs from various perspectives. The following are the major conclusions:

1) First, we assessed the present state of research and deployment of key SES technologies and outlined their current challenges. Current research on SESs is primarily concerned with six areas: uncertainty prediction, demand-side management, multi-energy flow coupling, ES technology, communication technology, and market strategy. We find that these technologies face the following challenges: a). Energy generation prediction in unanticipated extreme events. b). Door-to-door response techniques and price sensitivity diversification. c).

Online control of the dynamic characteristics of multi-energy flows. d). Large-scale parallel storage of multiple energy sources. e). The depth and breadth for applying communication technology and the digital energy market in a short period of time.

2) Then, considering the natural environment, technological constraints, infrastructure, and human existence from various angles, we examined the primary problems confronting the creation of SESs, for example: a). How to design an effective emergency response system capable of ensuring a stable energy supply in the event of a crisis? b). How to construct a network architecture capable of adapting flexibly to major calamities? c). How to address the tremendous energy waste problem? c). How to energetically expand society's available resources while conserving the environment. d). How to maintain a constant energy consumption level in the event of a major epidemic disaster? e). How to use carbon recovery technology in a large-scale and normalized manner? f). How to deal with unusual behaviors and sensitive visits throughout the information sharing process?
3) Finally, we assessed the significant opportunities that the SES will confront in the future from the perspectives of legislative support, technological innovation, and public participation. First, SES development must be linked with national development objectives and institutional changes to seize those opportunities. Then, the growth of SES must always be inextricably tied to the advancement of intelligent technology, focusing on breakthrough technologies that enable the energy system to acquire a high degree of intelligence. Lastly, SES development should encourage a more personalized and innovative service system, thereby expanding citizen participation.

6.2. Future works

To this aim, we propose that future development of SES should prioritize addressing current deficiencies, difficulties, and opportunities, as well as conducting divergent thinking and research from the following two main aspects:

1) On the one hand, future SES research should begin with a combination of technological innovation and practical application, maximizing the use of digital and intelligent technology and enabling the efficient and ubiquitous deployment of sophisticated technology in actual society. At this moment, the disconnect between frontier research on different critical technologies and their real unbalanced construction is one of the primary impediments to SES progress. As a result, creating of a robust infrastructure is a vital requirement for the evolution of various technologies inside SES. While pursuing technological innovation, it is necessary to enhance the performance of social infrastructure across the board to ensure widespread adoption and stable use of new technologies, achieve full integration of new technology development and application, and facilitate the natural transition of multi-energy systems to SES.
2) On the other hand, key technologies in the SES should consider the needs of society and people's livelihoods in various ways to progress more intelligently and diversely. For example, investigate a prediction technology capable of simultaneously predicting and analyzing multiple uncertainties in multiple subjects, propose a DR strategy based on accurate portrait positioning of users, investigate an ES technology that is scalable and capable of monitoring the operational status of SES in real time, or develop a rapid recovery and emergency avoidance technology for a system operating under weak stability and strong disturbance conditions, etc.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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